Part 1. Cobol

1. The author compares COBOL to asbestos because it was, “almost ubiquitous once upon a time and now incredibly, dangerously difficult to remove.” I believe the comparison makes sense.
2. The people who designed COBOL wanted it to be a long lasting easily maintainable code that does not need to be custom-written for specific machines. I think COBOL lives up to those expectations, since it is still used, and has lasted over a half century.
3. I think a programming language is successful if it is better than what came before it. Therefore COBOL is a successful programming language.
4. I think the AI tools mentioned in the last paragraph could become successful. AI can do the work faster than a human can. However, AI is not yet at the point to be trusted with all of the valuable information that it is going to be transferring. So I think AI will succeed, but not until humans perfect it for the task.

Part 3. Producing Errors

1. When producing the code 4 + [5, 6], a type error will occur because the code written is trying to use an operation that is not appropriate for the given object type.
2. When producing the code 2 + * 3, a syntax error will occur because the code written does not follow Python syntax.
3. When producing the code print(“Hello”), an indentation error will occur because the code is indented when it should not be.

Part 4. Definitions and Miscellany

1. Binary has two operands while unary has a single operand. Binary: (17 - 5) Unary: (-1)
2. Syntax is similar to how humans speak to each other where words can have different meanings depending on how you say them. Semantics on the other hand is a very literal language where both the words and their order need to be in a specific way to be understood.
3. Static typing is when a code cannot change its variables while it is being run. This means if x = 4 at the start, then x will still equal 4 at the end. Dynamic typing is when a code can change its variables while it is being run. This means if x = 4

at the start, then somewhere along the code x is changed to 8, then at the end it will be x = 8.

4. An integer is any whole number and the number 0. A float is when any number has a decimal point.

Part 6. Write a Homework Problem

Question:

Solve the distance between point A (1,7) and point B (12, 25) using Manhattan distance and solve the distance between points A and B using the Haversine formula. What is the difference between these two distances?

Problem Explanation:

This problem is meant to pick up where we left off in class. The person answering has to correctly use functions in their code to calculate the distances between the two points using two different distances. Then to connect the two distances, the difference between them is calculated. I think building the function, specifically using the Haversine formula, will be the most difficult part of the problem for the average student. Hopefully the student will become more familiar with functions and using functions in code.

Collaboration Log

I collaborated with Ben King on question 5.